ISYS1087/3478 Assignment 3 Group 50

JMeter Performance Analysis Report

Configuration 1: 1 virtual users result

Virtual Users	# Samples	Average	Median	90%	95%	99%	Min	Max	Error%	Throughput/sec
1	65	268	213	417	422	447	199	747	0.00	3.3

Configuration 2: 10 virtual users result

Virtual Users	# Samples	Average	Median	90%	95%	99%	Min	Max	Error%	Throughput/sec
10	2192	28	18	23	25	370	12	564	96.26	111.4

Configuration 3: 20 virtual users result

Virtual Users	# Samples	Average	Median	90%	95%	99%	Min	Max	Error%	Throughput/sec
20	4714	22	18	22	23	206	9	575	98.37	239.0

Configuration 4: 50 virtual users result

Virtual Users	# Samples	Average	Median	90%	95%	99%	Min	Max	Error%	Throughput/sec
50	12154	19	18	22	23	26	9	419	99.67	612.2

Analysis

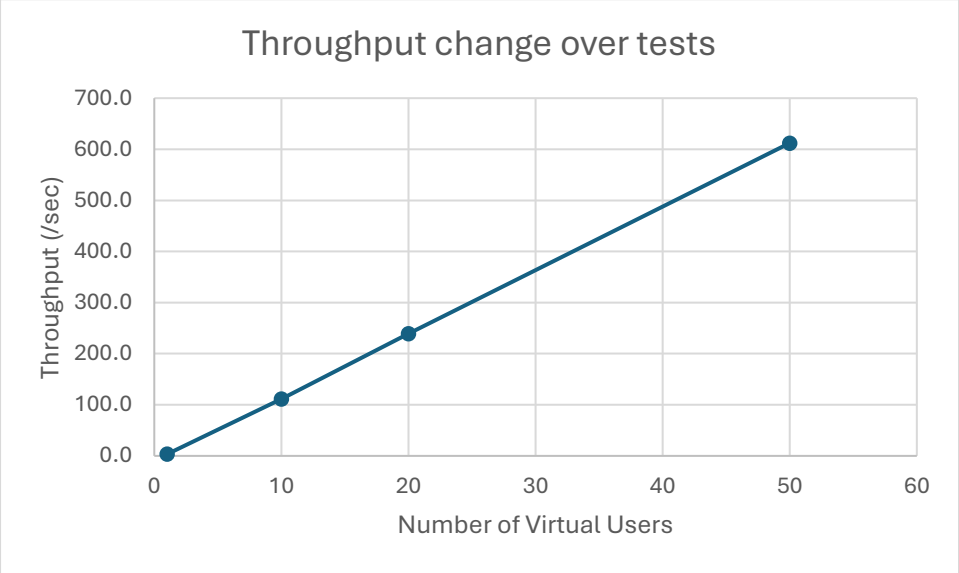
Semi-executive summary:

If the data were valid, then the change in results largely reflects diminishing returns as an increase of the virtual users and thus available samples provides more datapoints to make accurate measurements of, however some datapoints reflect a linear change in each successive test suite. These linear changes and diminishing returns all point towards an increase of efficiency as more tests are added, backed up by the throughput, minimum response time, average, and 95th percentile changing in a favourable direction.

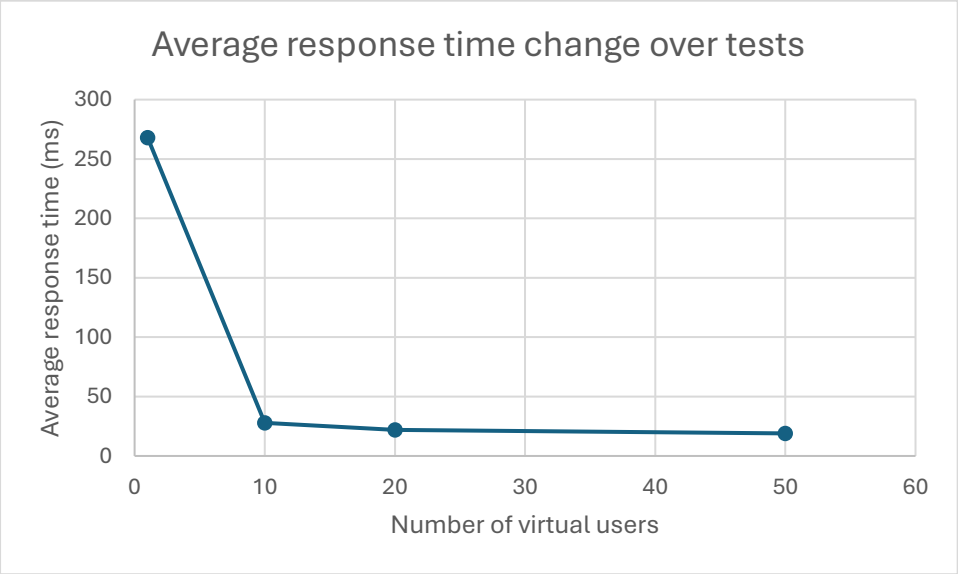
However, these results cannot be trusted and must be verified with alternative tests due to an issue with tests for 10, 20, and 50 virtual users being rate limited, and only the one virtual user being completed without error.

Individual metrics analysis

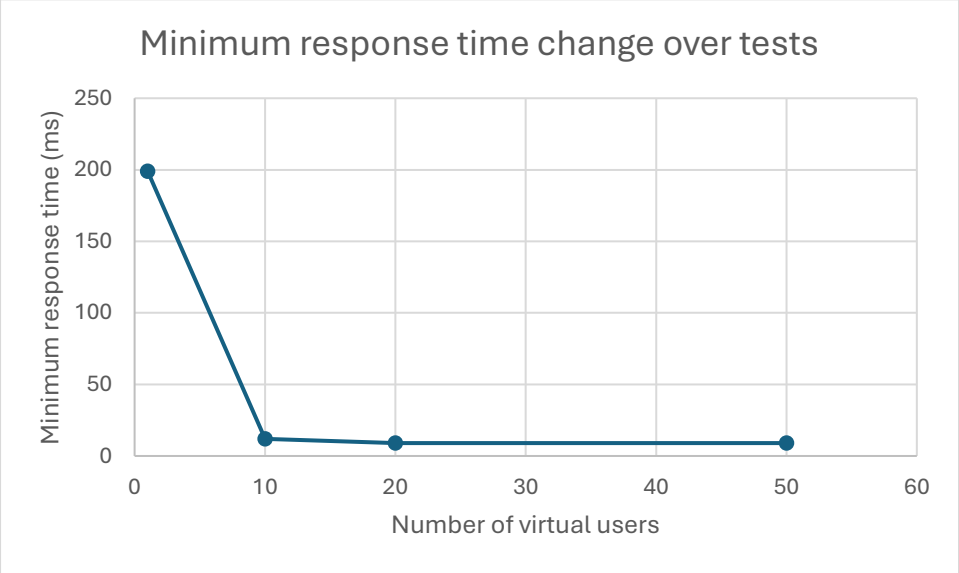
Throughput: As each successive suite of virtual users significantly increases the number of samples and tests, naturally the number of requests available to be processed increases. The linear growth is beneficial as being able to process more requests per minute means the tests can be completed in a shorter time, or there is suitable 'spare' time to increase the length of the tests or add more loops.



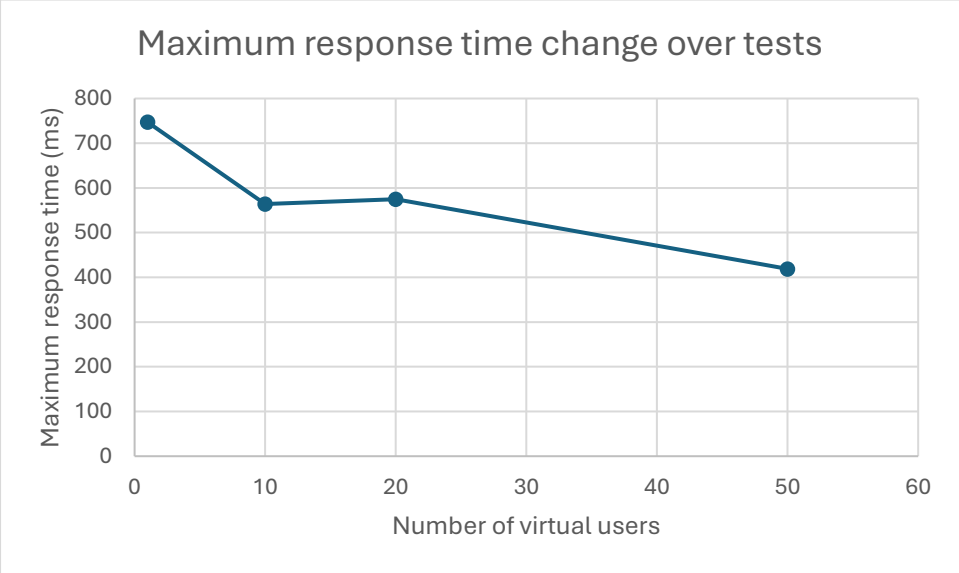
Average response time: The test with one virtual user reflects a relatively fast response time of 268ms then drops sharply from 10 virtual users onwards which is likely caused by the tests for 10, 20, and 50 virtual users being rate limited almost instantly, causing the JMeter tests to fail instantly, causing a 'false' diminishing returns. This is backed up by the error rate for one virtual user being 0% but climbs to 96%+ for the remaining tests.



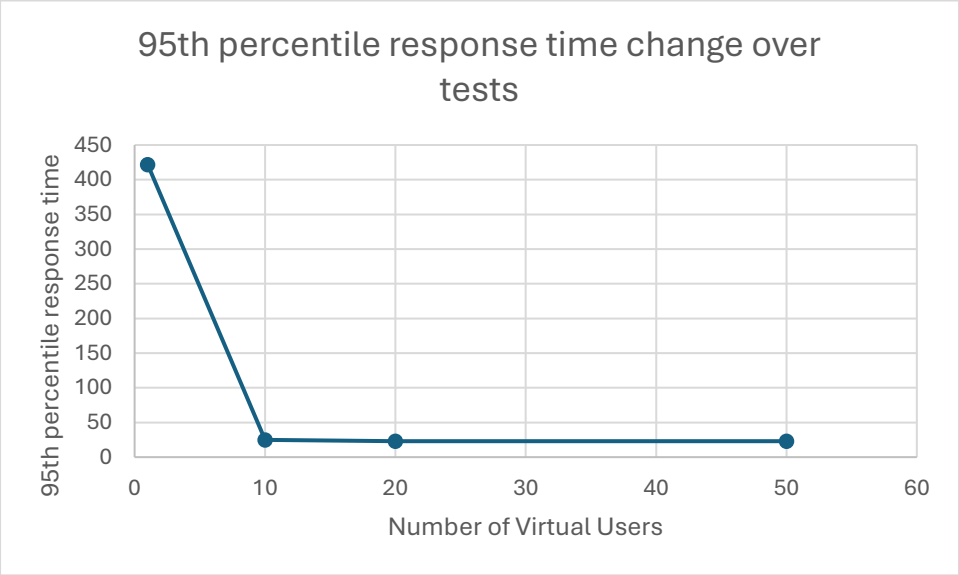
Minimum response time: Due to the same rate limiting error as discussed in ‘average response time’, the results shown here reflect a false diminishing return, where only a portion of the data can be trusted (one virtual user) and the remainder being unreliable. As the tests for 10, 20, and 50 virtual users suffered from rate limiting, ideally every test should be modified to avoid rate limiting and thus provide useful data.
If the minimum response times were valid, then the data suggests a gain in efficiency as more tests are added.



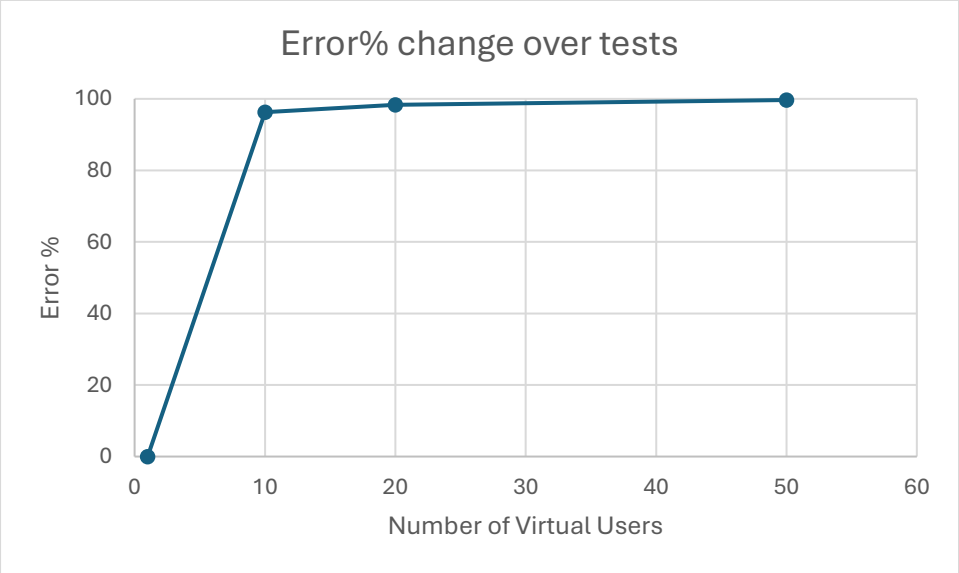
Maximum response time: As the rate limiting issue persists, the maximum response time also falls over each successive test, albeit linearly this time. The linear decrease can be explained by the increasing error rate, or the maximum response time only measuring the slowest recorded sample regardless of returned result.
If the data were to be treated as valid, then a decrease in maximum response time could reflect a change in how the server is handling the requests or be a normal variation in response times.



95th percentile response time: The values drops sharply from 1 to 10 virtual users due to the significant increase in samples available to compute data from (65 -> 2192), but adding further virtual users doesn't reflect the same decrease and only drops slightly, which shows diminishing returns. The reason for this may be explained by the aforementioned rate limiting issue.



Error %: The percentage jumps rapidly from 1 to 10 virtual users, 0% to 96.26%. These significant increase in error percentage is due to the JMeter tests for 10, 20, and 50 virtual users being rate limited by the ParaBank website, where it then only slightly increases due the increase of virtual users and samples only adding further errors without any passes or fails. This fast then slow growth reflects diminishing returns.



Appendix A: JMeter aggregate report data:

1 virtual user

Label ↑	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Throughput	Received KB/sec	Sent KB/sec
Accounts Overview	8	210	208	218	220	220	201	220	0.00%	27.6/min	2.75	0.09
Get Customer Acc...	8	212	212	218	218	218	208	218	0.00%	27.6/min	0.17	0.10
Login	8	422	417	431	447	447	413	447	0.00%	27.3/min	2.85	0.23
Logout	8	453	412	415	747	747	409	747	0.00%	27.3/min	2.19	0.18
Open Homepage	9	213	210	213	252	252	199	252	0.00%	27.2/min	2.09	0.08
Transfer Funds	8	214	216	218	218	218	209	218	0.00%	27.6/min	0.14	0.16
View Account Det...	8	211	209	218	220	220	205	220	0.00%	27.6/min	4.37	0.09
View Transactions	8	213	209	212	242	242	207	242	0.00%	27.6/min	4.13	0.11
TOTAL	65	268	213	417	422	447	199	747	0.00%	3.3/sec	16.68	0.93

10 virtual users

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Throughput	Received KB/sec	Sent KB/sec
Open Homepage	277	30	18	24	63	264	13	324	94.95%	14.1/sec	9.86	2.50
Login	276	36	18	22	25	413	13	557	95.65%	14.2/sec	10.16	4.33
Accounts Overview	274	26	18	22	24	209	12	372	95.99%	14.5/sec	9.87	2.63
Get Customer Acc...	274	25	18	23	25	203	12	206	95.99%	14.6/sec	6.68	3.16
View Account Det...	273	28	18	23	25	316	12	389	96.34%	14.7/sec	11.68	2.86
Transfer Funds	273	25	18	22	25	207	14	218	96.70%	15.0/sec	6.83	5.00
View Transactions	273	24	18	22	24	206	13	552	97.07%	15.1/sec	11.17	3.32
Logout	272	29	18	22	24	416	12	564	97.43%	15.3/sec	8.76	2.83
TOTAL	2192	28	18	23	25	370	12	564	96.26%	111.4/sec	71.21	25.23

20 virtual users

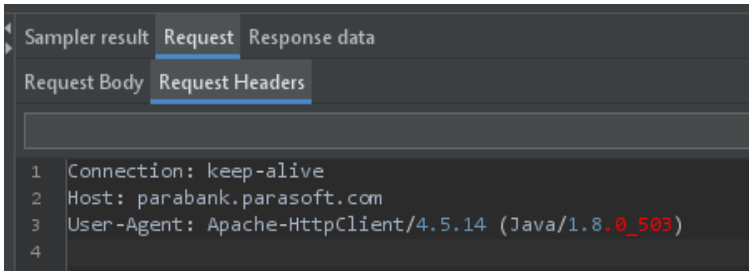
Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Throughput	Received KB/sec	Sent KB/sec
Open Homepage	596	23	18	22	24	248	9	276	97.82%	30.2/sec	17.25	5.00
Login	594	26	18	22	24	404	13	575	98.15%	30.5/sec	17.37	8.78
Accounts Overview	590	21	18	22	23	204	9	354	98.14%	31.0/sec	17.46	5.25
Get Customer Acc...	589	21	18	22	23	204	9	211	98.30%	31.2/sec	14.35	6.39
View Account Det...	587	21	17	22	23	204	11	372	98.47%	31.5/sec	18.87	5.73
Transfer Funds	586	21	18	22	23	203	13	211	98.46%	31.9/sec	14.62	10.26
View Transactions	586	20	18	21	22	203	11	300	98.81%	32.2/sec	18.87	6.69
Logout	586	22	17	21	22	394	9	554	98.81%	32.6/sec	16.70	5.53
TOTAL	4714	22	18	22	23	206	9	575	98.37%	239.0/sec	129.10	51.04

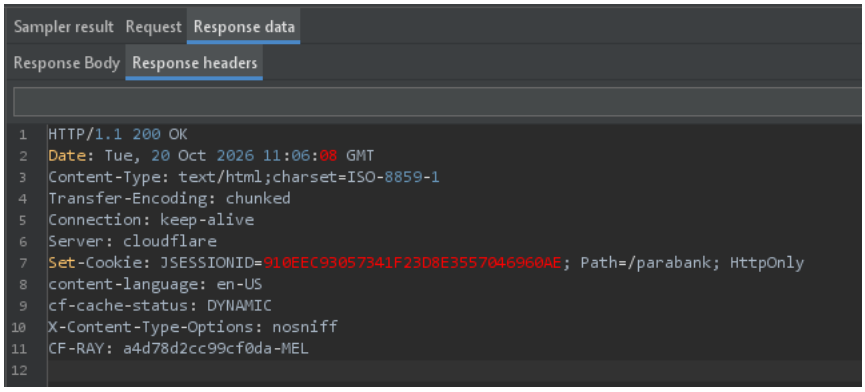
50 virtual users

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Throughput	Received KB/sec	Sent KB/sec
Open Homepage	1536	21	19	22	24	67	10	265	99.35%	77.9/sec	38.72	11.91
Login	1533	20	18	22	23	26	10	419	99.48%	78.8/sec	38.72	21.40
Accounts Overview	1526	19	18	22	23	25	9	209	99.54%	80.2/sec	38.99	12.53
Get Customer Acc...	1523	19	18	22	23	24	12	216	99.61%	80.9/sec	37.24	15.47
View Account Det...	1520	19	18	22	23	25	10	207	99.74%	81.0/sec	39.27	13.68
Transfer Funds	1508	18	18	22	23	25	11	211	99.80%	81.8/sec	37.70	25.26
View Transactions	1505	18	18	22	23	25	10	210	99.87%	82.6/sec	39.36	16.05
Logout	1503	18	18	21	22	24	10	226	100.00%	83.4/sec	38.49	12.89
TOTAL	12154	19	18	22	23	26	9	419	99.67%	612.2/sec	292.22	122.28

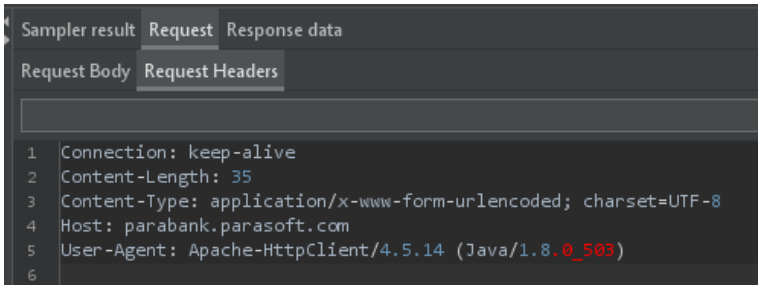
Appendix B: Request and Response Headers (not same tests as above):

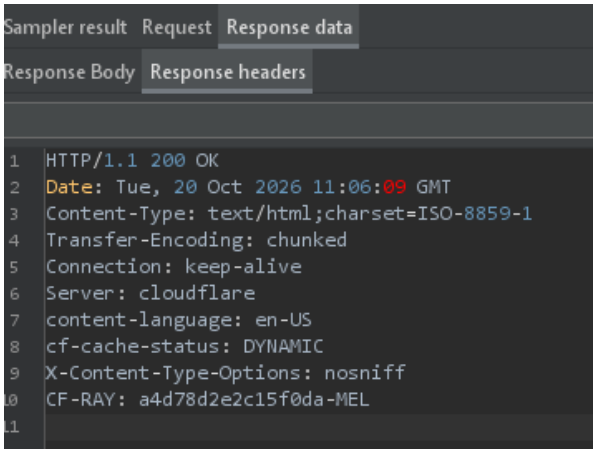
Open Homepage:



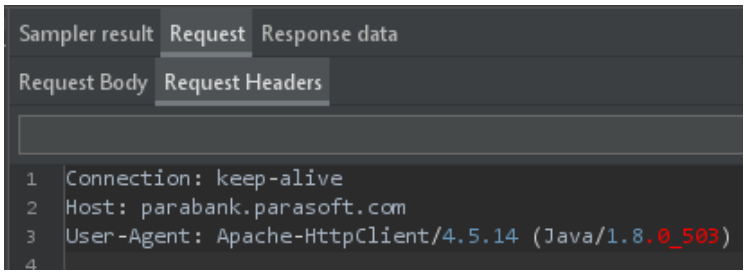


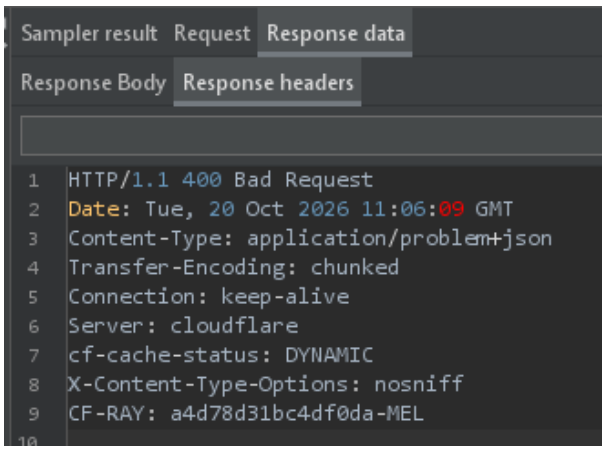
Login:



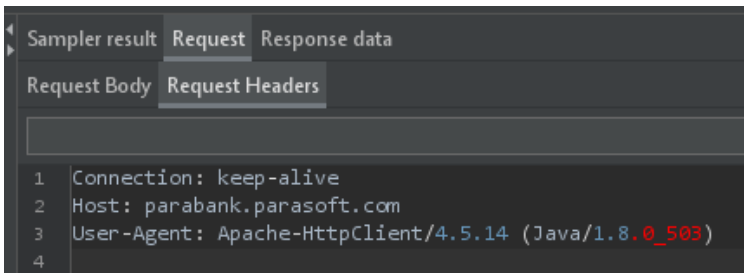


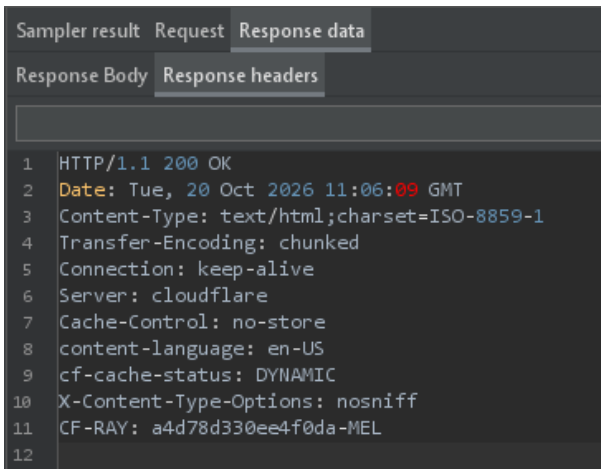
Accounts overview:





View account details:





Transfer funds:

Sampler result | Request | Response data

Request Body | Request Headers

```
1  Connection: keep-alive
2  Content-Length: 55
3  Content-Type: application/x-www-form-urlencoded; charset=UTF-8
4  Host: parabank.parasoft.com
5  User-Agent: Apache-HttpClient/4.5.14 (Java/1.8.0_503)
6
```

Sampler result | Request | Response data

Response Body | Response headers

```
1  HTTP/1.1 400 Bad Request
2  Date: Tue, 20 Oct 2026 11:06:10 GMT
3  Content-Type: application/problem+json
4  Transfer-Encoding: chunked
5  Connection: keep-alive
6  Server: cloudflare
7  cf-cache-status: DYNAMIC
8  X-Content-Type-Options: nosniff
9  CF-RAY: a4d78d3449e5f0da-MEL
10
```

View Transactions:

Sampler result | Request | Response data

Request Body | Request Headers

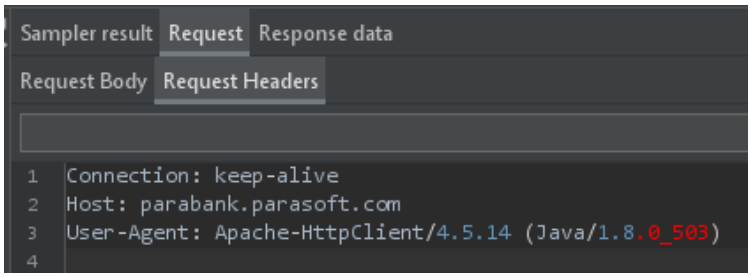
```
1  Connection: keep-alive
2  Host: parabank.parasoft.com
3  User-Agent: Apache-HttpClient/4.5.14 (Java/1.8.0_503)
4
```

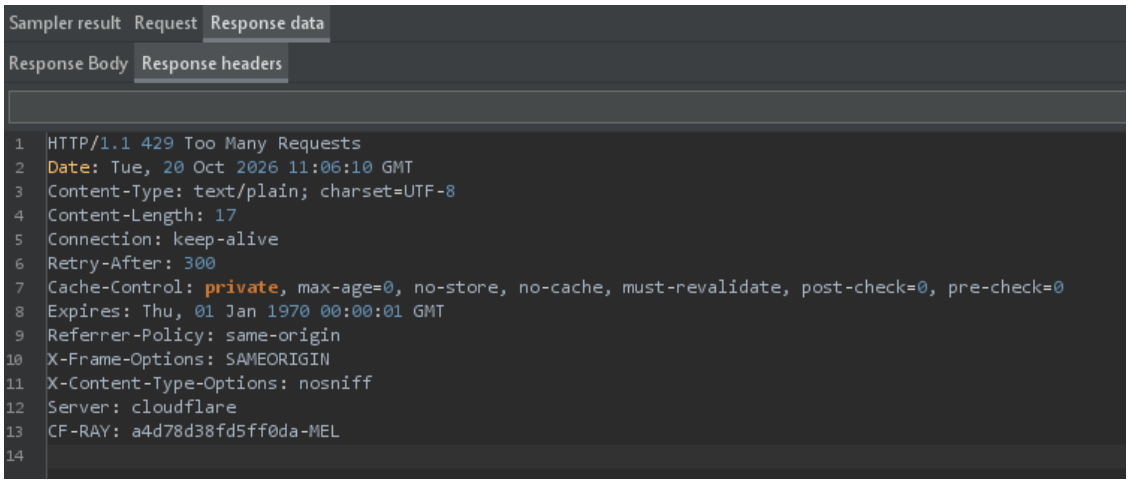
Sampler result | Request | Response data

Response Body | Response headers

```
1  HTTP/1.1 400 Bad Request
2  Date: Tue, 20 Oct 2026 11:06:10 GMT
3  Content-Type: application/problem+json
4  Transfer-Encoding: chunked
5  Connection: keep-alive
6  Server: cloudflare
7  cf-cache-status: DYNAMIC
8  X-Content-Type-Options: nosniff
9  CF-RAY: a4d78d367f09f0da-MEL
10
```

Logout:


Sampler result
Request
Response data
Request Body
Request Headers
1 Connection: keep-alive
2 Host: parabank.parasoft.com
3 User-Agent: Apache-HttpClient/4.5.14 (Java/1.8.0_503)
4


Sampler result
Request
Response data
Response Body
Response headers
1 HTTP/1.1 429 Too Many Requests
2 Date: Tue, 20 Oct 2026 11:06:10 GMT
3 Content-Type: text/plain; charset=UTF-8
4 Content-Length: 17
5 Connection: keep-alive
6 Retry-After: 300
7 Cache-Control: private, max-age=0, no-store, no-cache, must-revalidate, post-check=0, pre-check=0
8 Expires: Thu, 01 Jan 1970 00:00:01 GMT
9 Referrer-Policy: same-origin
10 X-Frame-Options: SAMEORIGIN
11 X-Content-Type-Options: nosniff
12 Server: cloudflare
13 CF-RAY: a4d78d38fd5ff0da-MEL
14